CNC Machine



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: CNC Machine

What it is used for

A CNC machine uses computer-controlled motion and tooling to machine parts to specified dimensions through operations such as milling, turning, drilling, or cutting. Follow the approved program, tooling setup, workholding method, and machine limits to achieve accurate parts while preventing collision, tool breakage, and serious injury.

Main parts

- Machine frame and work table or chuck
- Spindle and tool holder or turret
- Cutting tools and tool magazine, if fitted
- Axis drives, guides, and feed system
- CNC controller and operator panel
- Coolant or lubrication system, enclosure, and safety interlocks

Before you start

1. Confirm the machine type, material, drawing, tolerances, and approved machining program.
2. Inspect the enclosure, doors, interlocks, emergency stop, workholding, tools, and coolant system for damage.
3. Verify that the correct workpiece, fixtures, tool numbers, offsets, and program revision are selected.
4. Check that the workpiece and tools are securely clamped and that tool clearance has been verified.
5. Confirm coolant, lubrication, air supply, and chip-removal systems are ready where required.
6. Wear safety glasses and suitable footwear; secure long hair and remove loose clothing and jewellery.

How to use

1. Switch on the machine using the manufacturer's startup sequence and check the controller for active alarms.
2. Reference or home the axes using the approved procedure if required by the machine.
3. Load or select the authorised program and verify the program number, work coordinate system, and tool offsets.
4. Secure the workpiece and verify that clamps, chuck jaws, and fixtures cannot interfere with machine travel.

5. Check tool condition, tool length or radius offsets, spindle limits, and the intended cutting parameters.
6. Use simulation, graphics, or approved dry-run mode where available to verify the tool path and clearance.
7. Close the enclosure and confirm that all guards and interlocks are engaged before starting automatic motion.
8. Start the cycle and monitor the initial movements from the designated operator position.
9. Observe spindle load, coolant flow, cutting sound, chip formation, and any alarm indications.
10. Stop the cycle using the approved control if vibration, collision risk, tool breakage, or abnormal operation occurs.
11. Wait for all motion to stop before opening the enclosure or measuring the workpiece; isolate energy before reaching into hazardous areas.
12. Inspect the finished part against the drawing and record dimensions, offsets, tool changes, and quality results.
13. At completion, stop the machine, remove chips using approved tools, and leave the machine in the specified safe state.

⚠ Safety

- Only trained and authorised operators should set up, program, or operate the CNC machine.
- Never reach into the work area while the spindle or axes are moving; do not bypass door interlocks.
- Keep hands, clothing, hair, and measuring tools away from rotating tools and chucks.
- Use the correct rated tooling and workholding; never exceed machine or tool limits.
- Do not remove chips by hand or use compressed air in a way that can propel chips toward people.
- Apply lockout and tagout procedures before maintenance, tool changes requiring hazardous access, or clearing trapped material.

Care & storage

- Clean chips and residue from the work table, enclosure, and chip tray after safe shutdown.
- Inspect tools, holders, chuck or fixtures, guards, and coolant filters regularly.
- Check lubrication levels, coolant concentration, and fluid condition according to the maintenance schedule.
- Back up approved programs and record changes to tools, offsets, and machine settings.
- Arrange qualified service for repeated alarms, axis errors, abnormal vibration, or faulty safety interlocks.

Common mistakes to avoid

- Running an unverified program or incorrect work coordinate or tool offset.
- Starting automatic motion with unsecured workholding or incorrect tooling.
- Opening the enclosure or measuring a part before all motion has stopped.
- Ignoring alarms, tool wear, coolant failure, or dimensional deviations.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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